Year Up United
SQL Database Fundamentals, Part 2

Week 3 Lab Workbook

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Module 1

Querying Data

Exercise 1.A Getting Ready for Week 3

Getting Ready

This week you will continue to use git and GitHub to save your lab work as you write more complex SQL queries. You will also continue working with MySQL Workbench.

1. Save all homework to DataAnalytics/week-03.

2. You will need to have the Northwind sample database added to MySQL Workbench. If you have not yet completed Exercise 4.C from last week ("Adding the Northwind Database"), do so now.

Bonus Labs

The exercises in this workbook are your first priority! But if you find that you have time for additional exploration of SQL, the following **bonus labs** are recommended.

1. **SQLBolt** (https://sqlbolt.com/) has some nice tutorial and practice exercises that build up to more complex situations. If you have extra time or would like some additional practice understanding query logic, go there and start working your way through the exercises.

2. Similarly, **W3Schools** (https://www.w3schools.com/) has a helpful SQL tutorial with explanations of database concepts and keyword commands. Because the lessons are structured like a glossary of terms, this content also makes for a great browse-able general reference when you are trying to figure out what a keyword means or how to structure a query.

3. For some problem-solving practice, test out your developing SQL skills by solving the **SQL Murder Mystery**, https://mystery.knightlab.com/ 🕵️‍♀️

Exercise 1.B Querying a database

Answer the below questions by querying the Northwind database, saving your script to DataAnalytics/week-03 as **1B_query_practice.sql**. These questions use concepts from Module 1 of the Student Guide.

> **Tip #1**: Start your script with the query `USE northwind;` to specify what data is being queried.
>
> **Tip #2**: As you work, remember that you can "comment out" portions of your script, either by inserting `--` before the text, or by highlighting then using the keyboard shortcut **Ctrl /** to turn the highlighted text into a comment. Hitting **Ctrl /** again will remove the comment markup. This can be helpful as you work, e.g. to preserve a first attempt of a query for comparison while you rewrite and try a new version.
>
> **Tip #3**: You don't need to run the whole script every time you add a new query. Instead highlight just the portion you want to run and click the lightning bolt icon. (Or use the keyboard shortcut **Ctrl Enter** to run just the statement where your cursor is currently placed.)

Copy the text of each question below into your script as a `-- comment` to keep track of which queries are written in answer to each question.

1. Write a query to list the product id, product name, and unit price of every product that Northwind sells. (Hint: To help set up your query, look at the schema preview to see what column names belong to each table. Or use `SELECT *` to query all columns first, then refine your query to just the columns you want.)

2. Write a query to identify the products where the unit price is $7.50 or less.

3. What are the products that we carry where we have no units on hand, but 1 or more units are on backorder? Write a query that answers this question.

4. Examine the products table. How does it identify the type (category) of each item sold? Where can you find a list of all categories? Write a set of queries to answer these questions, ending with a query that creates a list of all the seafood items we carry.

5. Examine the products table again. How do you know what supplier each product comes from? Where can you find info on suppliers? Write a set of queries to find the specific identifier for "Tokyo Traders" and then find all products from that supplier.

6. How many employees work at northwind? What employees have "manager" somewhere in their job title? Write queries to answer each question.

7. Save your changes to 1B_query_practice.sql and use Git Bash to add, commit, and push to DataAnalytics/week-03.

Exercise 1.C More practice with queries

Continue answering the below questions by querying the Northwind database, saving your script to DataAnalytics/week-03 as **1C_query_practice.sql.**

Again, start your script with the query `USE northwind;` to specify what data is being queried.

Then copy the text of each question below into your script as a `-- comment` to keep track of which queries are written in answer to each question.

1. Write a query to list the product id, product name, and unit price of every product. This time, display them in ascending order by price.

2. What are the products that we carry where we have at least 100 units on hand? Order them in descending order by price.

3. What are the products that we carry where we have at least 100 units on hand? Order them in descending order by price. If two or more have the same price, list those in ascending order by product name.

4. Write a query against the orders table that displays the total number of distinct customers who have placed orders, based on customer ID. Use an alias to label the count calculation as CustomerCount.

5. Write a query against the orders table that displays the total number of distinct customers who have placed orders, by customer ID, for each country where orders have been shipped. Again, use an alias to label the count as CustomerCount. Order the list by the CustomerCount, largest to smallest.

6. What are the products that we carry where we have less than 25 units on hand, but 1 or more units of them are on order? Write a query that orders them by quantity on order (high to low), then by product name.

7. Write a query to list each of the job titles in employees, along with a count of how many employees hold each job title.

8. What employees have a monthly salary that is between $2000 and $2500? Write a query that orders them by job title.

9. Save your changes to 1C_query_practice.sql and use Git Bash to add, commit, and push to DataAnalytics/week-03.

Module 2

Querying Multiple Tables

Exercise 2.A Working with joins

Continue answering the below questions by querying the Northwind database, saving your script to DataAnalytics/week-03 as **2A_using_joins.sql**.

Again, start your script with the query `USE northwind;` to specify what data is being queried.

Then copy the text of each question below into your script as a `-- comment` to keep track of which queries are written in answer to each question.

1. Create a single query to list the product id, product name, unit price and category name of all products. Order by category name and within that, by product name.

2. Create a single query to list the product id, product name, unit price and supplier name of all products that cost more than $75. Order by product name.

3. Create a single query to list the product id, product name, unit price, category name, and supplier name of every product. Order by product name.

4. Create a single query to list the order id, ship name, ship address, and shipping company name of every order that shipped to Germany. Assign the shipping company name the alias 'Shipper.' Order by the name of the shipper, then the name of who it shipped to.

5. Start from the same query as above (#4), but omit OrderID and add logic to group by ship name, with a count of how many orders were shipped for that ship name.

6. Create a single query to list the order id, order date, ship name, ship address of all orders that included Sasquatch Ale.

   * Hint: You will need to join on three tables to accomplish this. (One of these tables has a sneaky space in the name, so you will need to surround it with backticks, like this: `table name`)

7. Save your changes to 2A_using_joins.sql and use Git Bash to add, commit, and push to DataAnalytics/week-03.

Module 3

Writing Complex Queries

Exercise 3.A Aggregate functions and grouping

Continue answering the below questions by querying the Northwind database, saving your script to DataAnalytics/week-03 as **3A_aggregations.sql**.

Again, start your script with the query `USE northwind;` to specify what data is being queried.

Then copy the text of each question below into your script as a `-- comment` to keep track of which queries are written in answer to each question.

1. Write a query to find the price of the cheapest item that Northwind sells. Then write a second query to find the name of the product that has that price.

2. Write a query to find the average price of all items that Northwind sells.
   (Bonus: Once you have written a working query, try asking Claude or ChatGPT for help using the ROUND function to round the average price to the nearest cent.)

3. Write a query to find the price of the most expensive item that Northwind sells. Then write a second query to find the name of the product with that price, plus the name of the supplier for that product.

4. Write a query to find total monthly payroll (the sum of all the employees' monthly salaries).

5. Write a query to identify the highest salary and the lowest salary amounts which any employee makes. (Just the amounts, not the specific employees!)

6. Write a query to find the name and supplier ID of each supplier and the number of items they supply. Hint: Join is your friend here.

7. Write a query to find the list of all category names and the average price for items in each category.

8. Write a query to find, for all suppliers that provide at least 5 items to Northwind, what is the name of each supplier and the number of items they supply.

9. Write a query to list products currently in inventory by the product id, product name, and inventory value (calculated by multiplying unit price by the number of units on hand). Sort the results in descending order by value. If two or more have the same value, order by product name. If a product is not in stock, leave it off the list.

10. Save your changes to 3A_aggregations.sql and use Git Bash to add, commit, and push to DataAnalytics/week-03.

Module 4

Using Subqueries

Exercise 4.A Working with subqueries

Continue answering the below questions by querying the Northwind database, saving your script to DataAnalytics/week-03 as **4A_subqueries.sql**.

Again, start your script with the query `USE northwind;` to specify what data is being queried.

Then copy the text of each question below into your script as a `-- comment` to keep track of which queries are written in answer to each question.

1. What is the product name(s) of the most expensive products?
   * HINT: Find the max price in a subquery and then use that value to find products whose price equals that value. (Some of your answers from Exercise 3.A may offer a useful starting point.)

2. What is the product name(s) and categories of the least expensive products?
   * HINT: Find the min price in a subquery and then use that in your more complex query that joins products with categories.

3. What is the order id, shipping name and shipping address of all orders shipped via "Federal Shipping"?
   * HINT: Find the shipper id of "Federal Shipping" in a subquery and then use that value to find the orders that used that shipper.
   * You do not need "Federal Shipping" to display in the results.

4. What are the order ids of the orders that included "Sasquatch Ale"?
   * HINT: Find the product id of "Sasquatch Ale" in a subquery and then use that value to find the matching orders from the `order details` table.
   * Your final results only need to display OrderID, but you may find it helpful to include additional columns as you work on creating the query to better understand how the query is working.

5. What is the name of the employee that sold order 10266?

6. What is the name of the customer that bought order 10266?

7. Save your changes to 4A_subqueries.sql and use Git Bash to add, commit, and push to DataAnalytics/week-03.

Reminder: Bonus Labs

Great work on reaching the end of the workbook exercises! With any time you have remaining this week, head back to Exercise 1.A and check out the bonus labs. ⚡ 🔎 🕵️